

# A correction to the proposed infinitude

$$y^3 - y = x^4 - x$$

An unconditional finiteness argument with its theorem dependence explicit

**Conclusion and scope.** The requested assertion is false. The equation has *finitely many* integer solutions. Below, the geometric calculation is given explicitly, and the arithmetic conclusion uses Siegel's theorem on integral points. That theorem is stated but not proved here: this is an unconditional proof, *not a fully self-contained arithmetic proof*. No proof of the requested infinitude can exist, and no complete enumeration of the solutions is claimed. No internet search was used.

**Theorem 1.** *The set*

$$\{(x, y) \in \mathbb{Z}^2 : y^3 - y = x^4 - x\}$$

*is finite. In fact, its projective algebraic curve is smooth and geometrically irreducible of genus 3.*

## 1. Intuition and elementary checks

The leading terms suggest  $y^3 \approx x^4$ , or  $y \approx |x|^{4/3}$ . This makes  $x = t^3$ ,  $y = t^4$  a natural first attempt. But substitution gives

$$(t^4)^3 - t^4 - ((t^3)^4 - t^3) = t^3(1 - t),$$

which vanishes only for  $t = 0, 1$ . Matching the leading powers does not match the lower terms. The following stronger observation makes this precise.

**Proposition 1.** *If  $(x, y)$  is an integer solution and  $|x|$  is a perfect cube, then  $x \in \{0, 1\}$  and  $y \in \{-1, 0, 1\}$ .*

*Proof.* Write  $f(s) = s^3 - s$ . For integers  $s \geq 1$ ,  $f(s+1) - f(s) = 3s(s+1) > 0$ . If  $x = \varepsilon t^3$ , where  $\varepsilon \in \{-1, 1\}$  and  $t \geq 2$ , put  $N = x^4 - x = t^{12} - \varepsilon t^3$ . Then

$$\begin{aligned} N - f(t^4) &= t^4 - \varepsilon t^3 > 0, \\ f(t^4 + 1) - N &= 3t^8 + 2t^4 + \varepsilon t^3 > 0. \end{aligned}$$

Thus  $N$  lies strictly between two consecutive values of  $f$  on the positive integers. Since  $N > 0$  and  $f(y) \leq 0$  for every integer  $y \leq 1$ , no integer  $y$  is possible. For  $x = -1$ ,  $N = 2$  lies between  $f(1) = 0$  and  $f(2) = 6$ . Finally,  $x = 0, 1$  gives  $y(y-1)(y+1) = 0$ .  $\square$

In particular, the six directly verified solutions are

$$(0, -1), (0, 0), (0, 1), (1, -1), (1, 0), (1, 1).$$

The proposition does not handle general  $x$ . To establish global finiteness, we next determine the geometry of the entire curve.

## 2. Irreducibility and absence of singularities

Let the affine curve be  $F(x, y) = 0$ , where

$$F(x, y) = y^3 - y - x^4 + x.$$

We work over  $\mathbb{C}$  so that the checks cover all possible components and singularities, including nonreal ones.

**Irreducibility.** As a cubic in  $y$  over the field  $\mathbb{C}(x)$ , a reducible  $F$  would have a root  $r(x) = p(x)/q(x)$ , with coprime polynomials  $p, q \in \mathbb{C}[x]$  and  $q \neq 0$ . Clearing denominators yields

$$p^3 - pq^2 = (x^4 - x)q^3.$$

Hence  $q$  divides  $p^3$ . Coprimality forces  $q$  to be constant, so  $r$  is a polynomial. A constant  $r$  cannot satisfy  $r^3 - r = x^4 - x$ ; a polynomial of degree  $d \geq 1$  would give  $3d = 4$ , also impossible. Thus  $F$  is irreducible over  $\mathbb{C}(x)$ , and, being monic in  $y$ , over  $\mathbb{C}[x, y]$  as well.

**Affine smoothness.** At an affine singularity, both partial derivatives must vanish:

$$F_x = 1 - 4x^3 = 0, \quad F_y = 3y^2 - 1 = 0.$$

Consequently,

$$x^3 = \frac{1}{4}, \quad y^2 = \frac{1}{3}.$$

Substituting  $x^4 = x/4$  and  $y^3 = y/3$  into  $F = 0$  gives

$$-\frac{2y}{3} + \frac{3x}{4} = 0, \quad y = \frac{9}{8}x, \quad x^2 = \frac{64}{243}.$$

But the two expressions for  $x^6$  would then require

$$\frac{1}{16} = \left(\frac{64}{243}\right)^3, \quad \text{hence} \quad 3^{15} = 2^{22},$$

which is impossible. There are no affine singularities.

**The point at infinity.** Homogenization gives the projective quartic

$$H(X, Y, Z) = Y^3Z - YZ^3 - X^4 + XZ^3 = 0.$$

On  $Z = 0$ , this requires  $X = 0$ , leaving exactly the point  $P_\infty = [0 : 1 : 0]$ . Since

$$H_Z = Y^3 - 3YZ^2 + 3XZ^2, \quad H_Z(P_\infty) = 1,$$

this point is smooth too. The homogenization is irreducible: any projective factorization would yield an affine one unless a factor were a power of  $Z$ , and  $Z$  does not divide  $H$ .

We have therefore obtained a smooth, geometrically irreducible projective curve  $C$ . Smoothness matters here: a singular quartic can have a much simpler underlying curve. The checks above exclude that possibility. The next page computes the genus directly, instead of merely quoting the genus formula for a smooth plane quartic.

### 3. A direct genus calculation and the finiteness step

The coordinate  $x$  defines a map  $C \rightarrow \mathbb{P}^1(\mathbb{C})$  of degree 3, since for general  $x$  the cubic equation has three distinct values of  $y$ . Geometric irreducibility makes the associated compact Riemann surface connected. A point has ramification index  $e$  when the map has local form  $t \mapsto t^e$  after local changes of coordinates.

**Finite ramification.** Where  $F_y \neq 0$ , the coordinate  $x$  is a local parameter, so there is no ramification. The remaining points have

$$y = \pm \frac{1}{\sqrt{3}}, \quad x^4 - x = -\frac{2y}{3}.$$

For each of these two  $y$ -values the quartic in  $x$  has four distinct roots: a repeated root would also have  $4x^3 - 1 = 0$ , giving an affine singularity already excluded. The two sets of  $x$ -values are disjoint. At each of these eight points, implicit differentiation with  $y$  as local parameter gives

$$\frac{dx}{dy} = 0, \quad \frac{d^2x}{dy^2} = \frac{6y}{4x^3 - 1} \neq 0.$$

Thus each has ramification index 2.

**Ramification at infinity.** In the chart  $Y = 1$ , put  $u = X/Y$ ,  $v = Z/Y$ . Near  $P_\infty$  the equation is

$$v - v^3 - u^4 + uv^3 = 0, \quad v(1 + (u - 1)v^2) = u^4.$$

Here  $u$  is a local parameter because the derivative with respect to  $v$  at  $(0, 0)$  is 1. The parenthesized factor is nonzero there, so  $v$  vanishes to order 4. Therefore  $x = u/v$  has a pole of order 3, and the ramification index at infinity is 3.

**Counting the genus.** Triangulate the sphere  $\mathbb{P}^1(\mathbb{C})$  with all branch values as vertices. Each open edge and face lifts three times. At a vertex, ramification of index  $e$  reduces the number of preimages by  $e - 1$  relative to the usual threefold count. The Euler characteristic is consequently

$$\chi(C) = 3\chi(\mathbb{P}^1) - \sum_P (e_P - 1) = 6 - (8(2 - 1) + (3 - 1)) = -4.$$

Since  $\chi(C) = 2 - 2g$ , the genus is  $g = 3$ .

**The arithmetic input (Siegel's theorem).** *If a geometrically irreducible affine algebraic curve over  $\mathbb{Q}$  has a smooth projective model of genus at least 1, then it has only finitely many integer points in its given affine coordinates.*

Our affine curve is defined over  $\mathbb{Q}$ , and the preceding computations give precisely such a model, of genus 3. Siegel's theorem therefore proves Theorem 1. This step has no unproved conjectural assumptions, but the theorem itself is a substantial external input.

**What is and is not established.** Finiteness is proved; infinitude is disproved. The six displayed points are verified, but are not proved to be the complete set. No explicit bound for all solutions, and no self-contained replacement for Siegel's theorem, is supplied here. These limitations do not change the finiteness conclusion.